Prime-Weighted Transfer for Centered Sieve Kernels: Large-Sieve Fixed Shifts and Power-Short Shift Means

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July 2026

Abstract

Let $Q = \prod_{w<p\le w^2} p$, let $G = (\mathbb{Z}/Q\mathbb{Z})^\times$, and let K_h° be the centered local kernel for the conditions $ab + h \not\equiv 0 \pmod p$, $p \mid Q$. We develop a weighted transfer theory for the associated factor-box form. The exact character expansion and fully active shift Parseval identity remain valid for arbitrary complex coefficients, while conductor localization permits two different analytic inputs.

First, exact-conductor orthogonality gives coefficient-uniform complete-shift cancellation. If I, J have lengths $M, N < Q$ and the coefficients have modulus at most one, then the normalized complete fully-active-shift mean square is

$$\ll \frac{1}{\max\{M,N\}} + \frac{1}{w \log w}.$$

The same asymptotic cancellation holds on every interval of $\lfloor Q^{1/\log w} \rfloor$ shift parameters, with an explicit deterministic completion error. Second, the classical multiplicative large sieve gives, uniformly for $(h, Q) = 1$, a fixed-shift estimate of the shape

$$|\mathcal{B}_h(a,b)| \ll \sqrt{E_a E_b} \left\{ \frac{\sqrt{MN}}{w} + (\log Q)(\sqrt{M} + \sqrt{N}) + \sqrt{Q} \right\},$$

where E_a, E_b are the coefficient energies on the Q-unit support. Thus arbitrary bounded coefficients have fixed-shift cancellation whenever the two lengths are power-sized and $MN/Q \to \infty$.

For von Mangoldt factor coefficients, we combine the same conductor decomposition with Siegel–Walfisz at small conductor. On dyadic boxes $I = (X, 2X]$, $J = (Y, 2Y]$ with $X = Q^{\lambda+o(1)}$ and $Y = Q^{\nu+o(1)}$, double von Mangoldt weights have complete and subpower-shift mean-square cancellation for every fixed $0 < \lambda, \nu < 1$. At a prescribed shift, including $h = 2$, the conclusion holds when $\lambda + \nu > 1$. Removing prime powers yields an asymptotic count of prime pairs (p, q) for which $pq + h$ avoids the primes in the window defining Q.

The target $pq + h$ is only required to be rough with respect to this finite window. We do not prove that it is prime, do not estimate $\Lambda(n)\Lambda(n + 2)$, and do not obtain a twin-prime lower bound. The ambient-volume condition $MN/Q \to \infty$ at a fixed shift is imposed by the present large-sieve argument and is not an asserted barrier.

Keywords: centered sieve kernel; multiplicative large sieve; von Mangoldt weight; Dirichlet character; fixed shift; rough product.

MSC 2020: Primary 11N35; Secondary 11L40, 11N13.

1 Introduction and scope

This paper makes a deliberately limited prime-weighted transfer. The input from the preceding papers is an exact finite character expansion for a centered Chinese-remainder pair kernel, including its conductor multipliers and a Parseval identity after averaging fully active shifts [18, 19]. The analytic inputs used here are classical: the multiplicative large sieve and the Siegel–Walfisz theorem. We assemble these ingredients first for arbitrary coefficients and then for von Mangoldt weights. We do not prove a new large sieve inequality, a new Siegel–Walfisz estimate, or a new theorem on primes in arithmetic progressions.

Let

$$Q = \prod_{w<p\le w^2} p, \qquad G = (\mathbb{Z}/Q\mathbb{Z})^\times, \qquad g = \varphi(Q), \qquad \kappa = \prod_{p\mid Q} \frac{p-2}{p-1}. \tag{1}$$

For intervals $I, J \subset \mathbb{Z}$ of respective lengths $M, N < Q$, coefficient sequences $a = (a_m)_{m\in I}$ and $b = (b_n)_{n\in J}$, and $(h, Q) = 1$, define

$$\mathcal{B}_h(a,b) = \sum_{\substack{m\in I,\ n\in J\\ (mn,Q)=1}} a_m b_n \left\{\kappa^{-1}\mathbf{1}_{(mn+h,Q)=1} - 1\right\}. \tag{2}$$

The centering in (2) is the exact complete-group centering. It is independent of the locations of the intervals and of the chosen coefficients. Put

$$E_a = \sum_{\substack{m\in I\\ (m,Q)=1}} |a_m|^2, \qquad E_b = \sum_{\substack{n\in J\\ (n,Q)=1}} |b_n|^2. \tag{3}$$

For $\chi \in \widehat{G}$, write q_χ for its primitive conductor and

$$A_{I,a}(\chi) = \sum_{\substack{m\in I\\ (m,Q)=1}} a_m\chi(m), \qquad A_{J,b}(\chi) = \sum_{\substack{n\in J\\ (n,Q)=1}} b_n\chi(n). \tag{4}$$

The exact kernel coefficient attached to a nonprincipal character has modulus

$$d(\chi) = \prod_{p\mid q_\chi} \frac{1}{p-2} \ll \frac{1}{q_\chi}. \tag{5}$$

This multiplier, rather than a generic estimate for a periodic kernel, is what permits conductor-by-conductor aggregation below.

1.1 Coefficient-uniform transfer

The first layer does not use the arithmetic nature of the coefficients. If $|a_m|, |b_n| \le 1$, exact-conductor orthogonality and the fully active shift Parseval identity give

$$\frac{1}{M^2N^2}\frac{1}{g}\sum_{\ell\in G} |\mathcal{B}_{2\ell}(a,b)|^2 \ll \frac{1}{\max\{M,N\}} + \frac{1}{w\log w}. \tag{6}$$

The estimate is uniform in the intervals and in all bounded coefficient sequences. It is an average over shifts 2ℓ for which every prime dividing Q remains active; it is not a conclusion for each ℓ. The exact identity and its coefficient-uniform consequence are stated in proposition 2.1 and theorem 2.5.

The complete shift mean can again be sampled on a subpower interval. If $\mathcal{L}$ is any interval of

$$H = \left\lfloor Q^{1/\log w} \right\rfloor \tag{7}$$

consecutive integers and $H_Q = \#\{\ell \in \mathcal{L} : (\ell, Q) = 1\}$, the deterministic completion argument transfers (6) to the average over those H_Q shifts. Although $H = Q^{o(1)}$, this remains a mean-square statement. In particular, it does not select a prescribed member of $\mathcal{L}$. See corollary 2.6.

At a fixed shift, the character phases no longer diagonalize. A dyadic conductor decomposition and the classical multiplicative large sieve yield the separate estimate

$$|\mathcal{B}_h(a,b)| \ll \sqrt{E_a E_b}\left\{\frac{\sqrt{MN}}{w} + (\log Q)(\sqrt{M}+\sqrt{N}) + \sqrt{Q}\right\}, \qquad (h,Q)=1. \tag{8}$$

The bound is uniform in the unit shift h. Consequently, bounded coefficients have normalized fixed-shift cancellation whenever M,N are fixed positive powers of Q and $MN/Q \to \infty$. A convenient uniform form is $M,N \geq Q^{\delta}$ and $MN \geq Q^{1+\delta}$ for fixed $\delta>0$. The ambient-volume threshold $MN=Q$ is the range furnished by this argument, not an asserted barrier. The precise uniform estimate is theorem 2.7.

The large-sieve mechanism in (8) is worth making explicit. On a dyadic conductor block $T < q_\chi \leq 2T$, the factor $d(\chi) \ll T^{-1}$ and the multiplicative large sieve control the weighted second moment of each transform. Cauchy–Schwarz is then applied within the same conductor block. Summing the resulting geometric blocks up to a cutoff and using full character Parseval above that cutoff gives (8); choosing the cutoff near $\sqrt{Q}$ balances the two sides. Thus the estimate is not a pointwise bound for an individual prime-supported character sum.

1.2 Double von Mangoldt weights

We next take dyadic factor intervals

$$I_X = (X,2X]\cap\mathbb{Z}, \qquad I_Y=(Y,2Y]\cap\mathbb{Z}, \qquad X,Y<Q, \tag{9}$$

and insert $a_m=\Lambda(m)$ and $b_n=\Lambda(n)$. Write the resulting form as $\mathcal{B}_h^{\Lambda,\Lambda}(X,Y)$. If

$$X = Q^{\lambda+o(1)}, \qquad Y=Q^{\nu+o(1)} \tag{10}$$

for fixed $0<\lambda,\nu<1$, then

$$\frac{1}{XY}\left(\frac{1}{g}\sum_{\ell\in G}|\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2\right)^{1/2} = o(1). \tag{11}$$

The same asymptotic conclusion holds on every shift interval of length (7). Hence all but $o(H_Q)$ of its unit shift parameters satisfy the corresponding $o(XY)$ estimate. This density-one conclusion still does not identify $\ell=1$ or any other prescribed shift. These statements are theorem 3.3 and corollaries 3.4 and 3.5.

The fixed-shift large-sieve estimate supplies a complementary result. For every $(h,Q)=1$, including $h=2$, we obtain

$$\mathcal{B}_h^{\Lambda,\Lambda}(X,Y) = o(XY) \qquad \text{when} \qquad \lambda+\nu>1. \tag{12}$$

Uniform versions are stated in terms of fixed lower power bounds for X,Y and a fixed power separation of XY from Q. The distinction between (11) and (12) is structural: shift orthogonality allows every pair of positive power lengths in the mean-square theorem, whereas the present fixed-shift large-sieve argument requires XY to exceed Q by a power; see theorem 3.6.

At small conductor, the prime-weighted input is the classical Siegel–Walfisz theorem. Uniformly for nonprincipal primitive characters of conductor at most a fixed power of $\log X$, it gives arbitrary logarithmic savings for the twisted summatory von Mangoldt function; see Siegel [14], Walfisz [17] and the modern formulation in Iwaniec and Kowalski [9, Corollary 5.29]. Restricting to $(n,Q)=1$ does not create a new prime-character-sum problem here. Since Λ is supported on prime powers, the discrepancy from the unrestricted primitive twist is bounded by

$$\sum_{p|Q}\sum_{\substack{k\geq 1\\ X<p^k\leq 2X}}\log p \leq \sum_{p|Q}\log p = \log Q, \tag{13}$$

because $p > 2$ makes a dyadic interval contain at most one power of each p. This is negligible on the power scales in (10). At complete and short shift means, larger conductors are controlled by the summable exact multiplier tail. At a prescribed shift, they are aggregated by the multiplicative large sieve of Montgomery and Vaughan [12]; standard accounts include Iwaniec and Kowalski [9, Chapter 7]. The elementary bounds $\sum_{n\le x}\Lambda(n) \ll x$ and $\sum_{n\le x}\Lambda(n)^2 \ll x\log x$ provide the required coefficient mass and energy. None of these analytic estimates is new here.

Removing prime powers from (12) gives a concrete prime-supported consequence. In its fixed-shift range,

$$\sum_{\substack{X<p\le 2X,\ Y<q\le 2Y\\ (pq+h,Q)=1}} (\log p)(\log q) = (\kappa + o(1))XY. \tag{14}$$

Uniform partial summation gives the corresponding unweighted asymptotic with main term $\kappa XY/(\log X \log Y)$. The condition on the target is exactly the one displayed: $pq + h$ avoids the primes in the finite window defining Q. The weighted and unweighted forms are recorded in corollary 3.7.

1.3 Relation to classical prime-distribution methods

The multiplicative large sieve is a classical mean-square inequality for arbitrary coefficients; see the foundational developments [1, 3, 6, 12]. Our use of it is specialized only through the exact multiplier (5) and the fact that exact local support equals primitive conductor for the squarefree modulus Q. We make no claim that the large-sieve inequality itself, or its application to a bare von Mangoldt transform, is new.

This is also not a new Bombieri–Vinogradov theorem. The classical theorem controls prime-distribution errors on average over a range of moduli [1, 16]; variants for other multiplicative sequences require their own hypotheses and exceptional-character analysis [7]. Here Q is one prescribed squarefree product. In (11) the average is over local shifts, not over moduli, while (12) is obtained from a weighted conductor decomposition internal to that single Q. These are different averaging geometries.

Vaughan's identity decomposes Λ into Type I and Type II pieces [15], but the identity alone estimates neither piece. The present proofs do not establish a new prime-sensitive Type II or dispersion estimate. Such estimates would become necessary if one replaced the local roughness indicator by a prime detector at the target, or tried to continue past the fixed-shift ranges supplied by (8). Likewise, results on prime-supported character sums in short progressions, such as Puchta [13], address a different averaging and parameter regime and are not silently imported here.

The large-modulus theorems of Bombieri et al. [2] and the fixed-residue and well-factorable results of Maynard [10, 11] use additional structure in the residue class, modulus family, or modulus weights. That structure is not supplied merely by factoring Q into primes or by observing that $d(q)$ is multiplicative. Those theorems are context for a possible future dispersion layer, not inputs that automatically strengthen the claims made here.

1.4 Strict boundary of the conclusions

The two factor variables in (14) are prime, but the target $pq + h$ is only Q-rough. In particular, even at $h = 2$ the theorem does not assert that $pq + 2$ is prime. It is not an estimate for

$$\sum_{n\le x}\Lambda(n)\Lambda(n+2),$$

and it gives no lower bound for twin primes. The shift-mean statements cannot be specialized to $h = 2$ without the independent fixed-shift theorem, and the latter still concerns a semiprime-type factor product followed by a finite roughness test.

Finally, local divisibility information does not remove the parity obstruction of sieve theory; see Heath-Brown [8] and the broader sieve framework in Friedlander and Iwaniec [5]. Prime-producing asymptotic sieves require further bilinear information absent here [4]. Accordingly, we do not describe (14) as evidence for the twin-prime conjecture and make no priority claim. The contribution is narrower: an exact CRT kernel converts classical Siegel–Walfisz and multiplicative large-sieve input into coefficient-uniform fixed-shift bounds and prime-weighted power-short shift means, with every remaining prime-detection interface left explicit.

2 Weighted kernel identities and large-sieve transfer

This section develops the coefficient-uniform part of the argument. All coefficients may be complex, and all interval estimates are uniform in the locations of the intervals. We first retain the exact finite Fourier identities, then use two different kinds of orthogonality. Orthogonality at one exact conductor controls complete shift means, while the classical multiplicative large sieve, summed over dyadic conductor blocks, controls a fixed shift in the range stated below.

2.1 The weighted centered form

Let $w \geq 5$, and put

$$\mathcal{P}(w) = \{p : w < p \leq w^2\}, \qquad Q = \prod_{p \in \mathcal{P}(w)} p, \qquad G = (\mathbb{Z}/Q\mathbb{Z})^\times, \qquad g = |G| = \varphi(Q). \tag{15}$$

Write $r = \omega(Q)$ and

$$\delta_Q = \frac{g}{Q}, \qquad \kappa = \prod_{p|Q} \frac{p-2}{p-1}, \qquad \Xi = \prod_{p|Q} \frac{p}{p-2}. \tag{16}$$

For an integer h satisfying $(h, Q) = 1$ and $a, b \in G$, define

$$K_h(a,b) = \prod_{p|Q} \mathbf{1}_{ab+h \not\equiv 0 \pmod p}, \tag{17}$$

$$K_h^\circ(a,b) = \kappa^{-1} K_h(a,b) - 1. \tag{18}$$

The factor κ is the exact row and column mean of K_h on G^2.

By the Chinese remainder theorem, every $\chi \in \widehat{G}$ has local components $\chi_p \in \widehat{\mathbb{F}_p^\times}$. Set

$$q_\chi = \prod_{\substack{p|Q \\ \chi_p \neq \mathbf{1}}} p, \qquad d(\chi) = d(q_\chi) = \prod_{p|q_\chi} \frac{1}{p-2}. \tag{19}$$

We extend characters by zero off the units. If $\chi \neq \mathbf{1}$, then it is induced by a primitive character of the squarefree conductor q_χ. Local character orthogonality gives

$$K_h^\circ(a,b) = \sum_{\substack{\chi \in \widehat{G} \\ \chi \neq \mathbf{1}}} c_h(\chi) \chi(a) \chi(b), \qquad c_h(\chi) = (-1)^{\omega(q_\chi)} d(\chi) \overline{\chi(-h)}. \tag{20}$$

In particular,

$$\#\{\chi \in \widehat{G} : q_\chi = q\} = \prod_{p|q} (p-2) = d(q)^{-1} \qquad (1 < q \mid Q). \tag{21}$$

Let $I, J \subset \mathbb{Z}$ be arbitrary integer intervals of respective lengths $1 \leq M, N < Q$. Let $(a_m)_{m \in I}$ and $(b_n)_{n \in J}$ be arbitrary complex coefficients. We use the effective unit-support norms

$$L_a = \sum_{\substack{m \in I \\ (m,Q)=1}} |a_m|, \qquad E_a = \sum_{\substack{m \in I \\ (m,Q)=1}} |a_m|^2, \tag{22}$$

$$L_b=\sum_{\substack{n\in J\\(n,Q)=1}}|b_n|,\qquad\qquad E_b=\sum_{\substack{n\in J\\(n,Q)=1}}|b_n|^2,\tag{23}$$

and the transforms

$$A_a(\chi)=\sum_{\substack{m\in I\\(m,Q)=1}}a_m\chi(m),\qquad\qquad A_b(\chi)=\sum_{\substack{n\in J\\(n,Q)=1}}b_n\chi(n).\tag{24}$$

The weighted centered box form is

$$\mathcal{B}_h(a,b)=\sum_{\substack{m\in I,\ n\in J\\(mn,Q)=1}}a_mb_nK_h^\circ(m,n).\tag{25}$$

Proposition 2.1 (Weighted character expansion and shift Parseval)**.** *For every* $(h,Q)=1$,

$$\mathcal{B}_h(a,b)=\sum_{\substack{\chi\in\widehat{G}\\ \chi\neq\mathbf{1}}}c_h(\chi)A_a(\chi)A_b(\chi).\tag{26}$$

Moreover, for the fully active shifts $h=2\ell$,

$$\frac{1}{g}\sum_{\ell\in G}|\mathcal{B}_{2\ell}(a,b)|^2=\sum_{\substack{\chi\in\widehat{G}\\ \chi\neq\mathbf{1}}}d(\chi)^2|A_a(\chi)|^2|A_b(\chi)|^2.\tag{27}$$

Both identities are exact.

Proof. For $p\nmid h$ and $t\in\mathbb{F}_p^\times$, orthogonality on $\mathbb{F}_p^\times$ gives

$$\frac{p-1}{p-2}\mathbf{1}_{t\neq-h}=1-\frac{1}{p-2}\sum_{\substack{\chi_p\in\widehat{\mathbb{F}_p^\times}\\ \chi_p\neq\mathbf{1}}}\overline{\chi_p(-h)}\chi_p(t).$$

Multiplying over $p\mid Q$ and deleting the all-principal term proves (20). Substitution into (25), followed by finite rearrangement, proves (26).

For $h=2\ell$,

$$c_{2\ell}(\chi)=(-1)^{\omega(q_\chi)}d(\chi)\overline{\chi(-2)}\,\overline{\chi(\ell)}.$$

After expanding the square in (26), the normalized average over $\ell\in G$ of $\overline{\chi(\ell)}\psi(\ell)$ is $\mathbf{1}_{\chi=\psi}$. This proves (27); complex coefficients cause no change because the second copy of the box form is conjugated. $\square$

2.2 Deterministic completion of a shift interval

Let $\mathcal{L}$ be an arbitrary interval of H consecutive integers and set

$$H_Q=\#\{\ell\in\mathcal{L}:(\ell,Q)=1\}.\tag{28}$$

Inclusion–exclusion gives

$$|H_Q-H\delta_Q|\leq 2^r.\tag{29}$$

Define the explicit completion cost

$$\mathfrak{C}(Q)=\kappa^{-2}4^r+2\kappa^{-1}3^r+2^r.\tag{30}$$

Proposition 2.2 (Weighted deterministic shift completion)**.** *For arbitrary complex coefficients as above,*

$$\left| \sum_{\substack{\ell\in\mathcal{L}\\(\ell,Q)=1}} |\mathcal{B}_{2\ell}(a,b)|^2 - \frac{H}{Q}\sum_{\ell\in G}|\mathcal{B}_{2\ell}(a,b)|^2 \right| \le \mathfrak{C}(Q)L_a^2L_b^2. \tag{31}$$

If $H\delta_Q \ge 2^{r+1}$, then

$$\frac{1}{H_Q}\sum_{\substack{\ell\in\mathcal{L}\\(\ell,Q)=1}} |\mathcal{B}_{2\ell}(a,b)|^2 \ll \frac{1}{g}\sum_{\ell\in G}|\mathcal{B}_{2\ell}(a,b)|^2 + \frac{\mathfrak{C}(Q)}{H\delta_Q}L_a^2L_b^2. \tag{32}$$

The implied constant is absolute.

Proof. Put

$$S_\ell = \sum_{\substack{m\in I,\ n\in J\\(mn,Q)=1}} a_m b_n K_{2\ell}(m,n), \qquad T = \left(\sum_{\substack{m\in I\\(m,Q)=1}} a_m\right)\left(\sum_{\substack{n\in J\\(n,Q)=1}} b_n\right).$$

Then $\mathcal{B}_{2\ell} = \kappa^{-1}S_\ell - T$. In the expansion of $|S_\ell|^2$, the unit condition on ℓ and the two kernel conditions exclude at most three residue classes modulo each $p \mid Q$. Expanding these exclusions by inclusion–exclusion produces at most 4^r terms. Each term is a single residue class modulo some divisor of Q, and its count in an interval differs from H/Q times its count in a complete period modulo Q by at most one. The corresponding discrepancies for $S_\ell\overline{T}$ and $|T|^2$ are at most 3^r and 2^r, respectively. The total absolute coefficient masses of all four terms are at most $L_a^2L_b^2$. The coefficients of the four terms in

$$|\kappa^{-1}S_\ell - T|^2 = \kappa^{-2}|S_\ell|^2 - \kappa^{-1}S_\ell\overline{T} - \kappa^{-1}\overline{S_\ell}T + |T|^2$$

therefore give exactly the right side of (31).

Under the stated lower bound for $H\delta_Q$, (29) implies $H_Q \ge H\delta_Q/2$. Divide (31) by H_Q and use $(H/Q)g = H\delta_Q$ to obtain (32). □

2.3 Exact-conductor orthogonality

The next lemma is a fixed-modulus orthogonality estimate, not a new large sieve inequality. Its advantage is that its modulus is the exact conductor q, rather than the ambient product Q.

Lemma 2.3 (Second moment at one exact conductor)**.** *For every divisor $q > 1$ of Q,*

$$\sum_{\substack{\chi\in\widehat{G}\\q_\chi=q}} |A_a(\chi)|^2 \le (M+q)E_a, \tag{33}$$

$$\sum_{\substack{\chi\in\widehat{G}\\q_\chi=q}} |A_b(\chi)|^2 \le (N+q)E_b. \tag{34}$$

The estimates are uniform in the interval locations.

Proof. Characters with $q_\chi = q$ are induced by primitive characters modulo q, so they form a subset of all Dirichlet characters modulo q. Put $u_m = a_m\mathbf{1}_{(m,Q)=1}$. On the support of u_m, the inflated character χ equals its inducing character modulo q. Orthogonality modulo q therefore gives

$$\sum_{\chi \bmod q} \left| \sum_{m\in I} u_m \chi(m) \right|^2 = \varphi(q) \sum_{\substack{c \bmod q \\ (c,q)=1}} \left| \sum_{\substack{m\in I \\ m\equiv c \pmod q}} u_m \right|^2 .$$

Every residue class contains at most $M/q+1$ members of I. Cauchy's inequality inside each class, followed by $\varphi(q)\le q$, bounds the last display by $(M+q)E_a$. Restricting the nonnegative sum to the exact conductor characters proves (33). The proof of (34) is identical. □

We record the conductor masses which occur after applying the preceding lemma. The final estimate, which removes the one-prime supports, is useful for prime-supported coefficients later in the paper.

Lemma 2.4 (Conductor masses in the quadratic window)**.** *As $w\to\infty$,*

$$\sum_{\substack{q\mid Q \\ q>1}} q\, d(q)^2 = O(1), \tag{35}$$

$$\sum_{\substack{q\mid Q \\ q>1}} d(q)^2 \ll \frac{1}{w\log w}, \tag{36}$$

$$\sum_{\substack{q\mid Q \\ \omega(q)\ge 2}} d(q)^2 \ll \frac{1}{w^2(\log w)^2}. \tag{37}$$

Moreover, $\delta_Q = 1/2 + o(1)$, $\kappa = 1/2 + o(1)$, and $\Xi = 4 + o(1)$.

Proof. Euler products give

$$\sum_{q\mid Q} q\, d(q)^2 = \prod_{p\mid Q}\left(1+\frac{p}{(p-2)^2}\right),$$

$$\sum_{q\mid Q} d(q)^2 = \prod_{p\mid Q}\left(1+\frac{1}{(p-2)^2}\right).$$

Mertens' theorem on $w<p\le w^2$ shows that $\sum_{p\mid Q} p/(p-2)^2 = O(1)$, proving (35). Also

$$s_w := \sum_{p\mid Q}\frac{1}{(p-2)^2} \ll \frac{1}{w\log w}.$$

Thus the nonempty part of the second Euler product is $O(s_w)$, which proves (36). The portion containing at least two local factors is at most $e^{s_w}-1-s_w = O(s_w^2)$, proving (37). The remaining local-product asymptotics follow from the prime number theorem and Mertens' theorem. □

Theorem 2.5 (Coefficient-uniform complete-shift mean square)**.** *For arbitrary complex coefficients,*

$$\frac{1}{g}\sum_{\ell\in G} |\mathcal{B}_{2\ell}(a,b)|^2 \le L_b^2 E_a \sum_{\substack{q\mid Q \\ q>1}} d(q)^2 (M+q), \tag{38}$$

$$\frac{1}{g}\sum_{\ell\in G} |\mathcal{B}_{2\ell}(a,b)|^2 \le L_a^2 E_b \sum_{\substack{q\mid Q \\ q>1}} d(q)^2 (N+q). \tag{39}$$

In particular, if $|a_m|, |b_n| \le 1$, then

$$\frac{1}{M^2N^2}\frac{1}{g}\sum_{\ell\in G}|\mathcal{B}_{2\ell}(a,b)|^2 \ll \frac{1}{\max\{M,N\}} + \frac{1}{w\log w}. \tag{40}$$

The implied constant is absolute for all sufficiently large w.

Proof. In (27), group the characters by their exact conductor and use $|A_b(\chi)| \le L_b$. Lemma 2.3 gives

$$\frac{1}{g}\sum_{\ell\in G}|\mathcal{B}_{2\ell}(a,b)|^2 \le L_b^2 E_a \sum_{\substack{q|Q\\ q>1}} d(q)^2(M+q),$$

which is (38). Interchanging the two variables proves (39).

For coefficients bounded by one, $L_a \le M$, $E_a \le M$, $L_b \le N$, and $E_b \le N$. The first inequality, divided by M^2N^2, is at most

$$\frac{1}{M}\sum_{\substack{q|Q\\ q>1}} q\, d(q)^2 + \sum_{\substack{q|Q\\ q>1}} d(q)^2.$$

The second inequality gives the same expression with N in place of M. Take the smaller of the two bounds and apply lemma 2.4. □

For the quadratic window, the prime number theorem gives

$$\log Q = (1+o(1))w^2, \qquad r = (1+o(1))\frac{w^2}{2\log w}. \tag{41}$$

Consequently, for

$$H = \left\lfloor Q^{1/\log w}\right\rfloor, \tag{42}$$

one has

$$\frac{\mathfrak{C}(Q)}{H\delta_Q} = \exp\left(-(1-\log 2+o(1))\frac{w^2}{\log w}\right) = o(1). \tag{43}$$

Corollary 2.6 (Coefficient-uniform short-shift mean square)**.** *Let $\mathcal{L}$ be any interval of H consecutive integers, with H as in (42). If $|a_m|, |b_n| \le 1$, then*

$$\frac{1}{M^2N^2}\frac{1}{H_Q}\sum_{\substack{\ell\in\mathcal{L}\\ (\ell,Q)=1}}|\mathcal{B}_{2\ell}(a,b)|^2 \ll \frac{1}{\max\{M,N\}} + \frac{1}{w\log w} + \frac{\mathfrak{C}(Q)}{H\delta_Q}. \tag{44}$$

In particular, the normalized root mean square is $o(1)$ whenever $\max\{M,N\}\to\infty$.

Proof. For large w, (43) implies $H\delta_Q \ge 2^{r+1}$. Apply (32), use $L_a \le M$ and $L_b \le N$, and then invoke theorem 2.5. □

2.4 A dyadic large-sieve estimate at a fixed shift

The preceding mean-square result exploits orthogonality in the shift. At a fixed shift we instead sum exact conductors in dyadic blocks. We use the classical multiplicative large sieve in the form

$$\sum_{q\le S}\frac{q}{\varphi(q)}\sideset{}{^*}\sum_{\chi \bmod q}\left|\sum_{n\in\mathcal{I}}u_n\chi(n)\right|^2 \ll (|\mathcal{I}|+S^2)\sum_{n\in\mathcal{I}}|u_n|^2, \tag{45}$$

where the star denotes primitive characters. The estimate is uniform in the location of the integer interval $\mathcal{I}$; see, for example, Iwaniec and Kowalski [9, Theorem 7.13]. We apply this classical result only to the sparse family of squarefree conductors dividing Q.

For $R \ge 1$, let $\mathcal{B}_h^{>R}(a,b)$ denote the portion of (26) with $q_\chi > R$.

Theorem 2.7 (Dyadic multiplicative-large-sieve bound)**.** *Let I, J be arbitrary integer intervals of lengths $M, N < Q$, let the coefficients be arbitrary complex numbers, and let $(h, Q) = 1$. Uniformly for $w \le R \le \sqrt{Q}$,*

$$|\mathcal{B}_h^{>R}(a,b)| \ll \sqrt{E_a E_b}\left\{\frac{\sqrt{MN}}{R} + (\log Q)(\sqrt{M}+\sqrt{N}) + \sqrt{Q}\right\}. \tag{46}$$

Since every nontrivial conductor q_χ is larger than w, the full form satisfies

$$|\mathcal{B}_h(a,b)| \ll \sqrt{E_a E_b}\left\{\frac{\sqrt{MN}}{w} + (\log Q)(\sqrt{M}+\sqrt{N}) + \sqrt{Q}\right\}. \tag{47}$$

All implied constants are absolute for sufficiently large w, and the estimates are uniform in h and in the interval locations.

Proof. Partition $R < q \le \sqrt{Q}$ into geometric blocks $S < q \le \min\{2S, \sqrt{Q}\}$, beginning with $S = R$. Since

$$d(q) = \frac{1}{q}\prod_{p\mid q}\frac{p}{p-2} \le \frac{\Xi}{q},$$

Cauchy's inequality gives

$$\begin{aligned}&\left|\sum_{\substack{q\mid Q\\ S<q\le\min\{2S,\sqrt{Q}\}}}\ \sum_{\substack{\chi\in\widehat{G}\\ q_\chi=q}} c_h(\chi)A_a(\chi)A_b(\chi)\right|\\ &\quad\le \frac{\Xi}{S}\left(\sum_{\substack{q\mid Q\\ S<q\le\min\{2S,\sqrt{Q}\}}}\ \sum_{q_\chi=q}|A_a(\chi)|^2\right)^{1/2}\left(\sum_{\substack{q\mid Q\\ S<q\le\min\{2S,\sqrt{Q}\}}}\ \sum_{q_\chi=q}|A_b(\chi)|^2\right)^{1/2}.\end{aligned}$$

Every exact-conductor character in these sums is primitive modulo q. Apply (45) to $a_m\mathbf{1}_{(m,Q)=1}$ and $b_n\mathbf{1}_{(n,Q)=1}$, enlarging the conductor family to all primitive characters of modulus at most $2S$. The block is at most

$$\begin{aligned}&\frac{\Xi}{S}\sqrt{(M+4S^2)(N+4S^2)}\sqrt{E_aE_b}\\ &\le \Xi\sqrt{E_aE_b}\left(\frac{\sqrt{MN}}{S} + 2\sqrt{M} + 2\sqrt{N} + 4S\right).\end{aligned} \tag{48}$$

Summing (48) over dyadic S from R to $\sqrt{Q}$ uses

$$\sum_S \frac{1}{S} \ll \frac{1}{R}, \qquad \#\{S\} \ll \log Q, \qquad \sum_S S \ll \sqrt{Q}.$$

It remains to treat $q_\chi > \sqrt{Q}$. Character Parseval on G gives

$$\sum_{\chi\in\widehat{G}}|A_a(\chi)|^2 = gE_a, \qquad \sum_{\chi\in\widehat{G}}|A_b(\chi)|^2 = gE_b. \tag{49}$$

Here the hypothesis $M, N < Q$ ensures that no two integers in either interval occupy the same residue class modulo Q. Since $d(\chi) \le \Xi/q_\chi \le \Xi/\sqrt{Q}$, Cauchy's inequality bounds this remaining portion by

$$\frac{\Xi}{\sqrt{Q}}\, g\sqrt{E_aE_b} \le \Xi\sqrt{Q}\sqrt{E_aE_b}.$$

Combining the two conductor ranges and using $\Xi = O(1)$ proves (46). Finally, all primes dividing Q exceed w, so every $q_\chi > 1$ exceeds w. The choice $R = w$ gives (47). □

Corollary 2.8 (A fixed-shift cancellation range)**.** *If* $|a_m|, |b_n| \le 1$*, then, uniformly for* $(h, Q) = 1$*,*

$$\frac{|\mathcal{B}_h(a,b)|}{MN} \ll \frac{1}{w} + (\log Q)\left(\frac{1}{\sqrt{M}} + \frac{1}{\sqrt{N}}\right) + \sqrt{\frac{Q}{MN}}. \tag{50}$$

Consequently, if

$$\frac{\min\{M,N\}}{(\log Q)^2} \longrightarrow \infty, \qquad \frac{MN}{Q} \longrightarrow \infty, \tag{51}$$

then $\mathcal{B}_h(a,b) = o(MN)$*. In particular, for* $M = Q^{\lambda+o(1)}$ *and* $N = Q^{\nu+o(1)}$ *with fixed* $\lambda, \nu > 0$*, this holds whenever* $\lambda + \nu > 1$*.*

Proof. Under the coefficient bound, $E_a \le M$ and $E_b \le N$. Substitute these estimates in (47) and divide by MN. □

Remark 2.9 (Strict scope of the fixed-shift estimate). The estimate in theorem 2.7 is a bilinear large-sieve transfer for this particular centered CRT kernel. It requires the ambient-volume condition visible in (51); it is not a complete prime-sensitive Type II theorem. The shift-mean results retain their shift average, while the fixed-shift result retains its length restriction. Furthermore, K_h only tests whether a product plus h avoids the primes dividing Q. None of the results in this section tests whether that target is prime, estimates $\Lambda(n)\Lambda(n+2)$, overcomes sieve parity, or implies a twin-prime lower bound.

3 Prime-weighted applications

This section inserts von Mangoldt weights in both factor coordinates. The factor intervals are dyadic: throughout,

$$I_X = (X, 2X] \cap \mathbb{Z}, \qquad I_Y = (Y, 2Y] \cap \mathbb{Z}, \qquad 2 \le X, Y < Q. \tag{52}$$

The condition that the length of each interval is below Q will be used only when applying Parseval on G: no two integers in either interval then occupy the same residue class modulo Q.

For $\chi \in \widehat{G}$, define the Q-unit von Mangoldt transform

$$A_{\Lambda,X}(\chi) := \sum_{\substack{n \in I_X \\ (n,Q)=1}} \Lambda(n)\chi(n), \tag{53}$$

and define $A_{\Lambda,Y}$ analogously. The corresponding weighted centered box form is

$$\mathcal{B}_h^{\Lambda,\Lambda}(X,Y) := \sum_{\substack{m \in I_X,\ n \in I_Y \\ (mn,Q)=1}} \Lambda(m)\Lambda(n) K_h^{\circ}(m,n). \tag{54}$$

Thus, for $(h, Q) = 1$, the general weighted character expansion gives

$$\mathcal{B}_h^{\Lambda,\Lambda}(X,Y) = \sum_{\substack{\chi \in \widehat{G} \\ \chi \ne \mathbf{1}}} c_h(\chi) A_{\Lambda,X}(\chi) A_{\Lambda,Y}(\chi), \qquad |c_h(\chi)| = d(\chi). \tag{55}$$

3.1 Primitive comparison and small conductors

The unit restriction in (53) causes no loss at polylogarithmic conductor. The reason is special to Λ: a term removed by the restriction is a prime power, rather than a general multiple of a prime in the sieving window.

Lemma 3.1 (Primitive comparison for a dyadic Mangoldt sum). *Let $\chi \in \widehat{G}$ have exact conductor $q = q_\chi > 1$, and let χ^* be the primitive character modulo q inducing χ. Then*

$$A_{\Lambda,X}(\chi) = \sum_{X<n\le 2X} \Lambda(n)\chi^*(n) + O(\log Q), \tag{56}$$

uniformly in $X < Q$ and in χ. Consequently, fix $\delta > 0$ and $A, B > 0$. Uniformly for $Q^\delta \le X < Q$ and $1 < q_\chi \le w^A$,

$$|A_{\Lambda,X}(\chi)| \ll_{A,B,\delta} X(\log X)^{-B}. \tag{57}$$

Proof. On integers coprime to Q, the inflated character χ agrees with χ^*. If $\Lambda(n) \ne 0$ and $(n, Q) > 1$, then $n = p^k$ for a unique prime $p \mid Q$. When $p \mid q$, the value $\chi^*(p^k)$ already vanishes. When $p \mid Q/q$, the term is deleted by the Q-unit restriction. Since $p > 2$, the dyadic interval $(X, 2X]$ contains at most one power of a fixed p. The total weight of all possibly deleted terms is therefore at most

$$\sum_{p\mid Q} \log p = \log Q,$$

which proves (56).

Every character under consideration is primitive and nonprincipal. Moreover, when $X \ge Q^\delta$, one has $w^A \le (\log X)^{A/2+o(1)}$. The classical Siegel–Walfisz theorem, in its character form, gives an arbitrary fixed logarithmic saving uniformly in this range [9, 14, 17]. Increasing that saving if necessary absorbs the $O(\log Q)$ term and proves (57). □

As usual for an unconditional Siegel–Walfisz estimate, the implied constants in (57), and hence those in the prime-weighted $o(1)$ conclusions below, are ineffective.

We also need a tail estimate for the exact conductor mass. It is important here that the tail begins at a power of w, not at a power of Q.

Lemma 3.2 (Polylogarithmic conductor tail). *For every fixed $A > 0$,*

$$\sum_{\substack{q\mid Q\\ q>w^A}} d(q) \ll e^{-A}, \tag{58}$$

with an absolute implied constant for all sufficiently large w.

Proof. Take $\theta = 1/\log w$. Rankin's inequality gives

$$\begin{aligned}\sum_{\substack{q\mid Q\\ q>w^A}} d(q) &\le e^{-A}\sum_{q\mid Q} q^\theta d(q)\\ &= e^{-A}\prod_{p\mid Q}\left(1 + \frac{p^{1/\log w}}{p-2}\right).\end{aligned}$$

Since $w < p \le w^2$, the numerator in the local perturbation is at most e^2, and Mertens' theorem shows that the product is $O(1)$. □

3.2 Two prime weights after complete or short shift averaging

The complete-shift identity takes the exact form

$$\frac{1}{g}\sum_{\ell\in G} |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2 = \sum_{\substack{\chi\in\widehat{G}\\ \chi\ne\mathbf{1}}} d(\chi)^2 |A_{\Lambda,X}(\chi)|^2 |A_{\Lambda,Y}(\chi)|^2. \tag{59}$$

Theorem 3.3 (Two Mangoldt weights on every fixed power scale)**.** *Fix $0 < \delta < 1$. Uniformly for*

$$Q^{\delta} \leq X, Y < Q, \tag{60}$$

one has

$$\frac{1}{XY} \left(\frac{1}{g} \sum_{\ell \in G} |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2 \right)^{1/2} = o(1). \tag{61}$$

In particular, the conclusion holds when $X = Q^{\lambda + o(1)}$ and $Y = Q^{\nu + o(1)}$ for arbitrary fixed $0 < \lambda, \nu < 1$.

Proof. Fix $A > 0$ temporarily and split (59) at $q_\chi = w^A$. For the low-conductor part, lemma 3.1 and the identity

$$\sum_{q_\chi = q} d(\chi)^2 = d(q)$$

give, for any prescribed logarithmic saving,

$$\frac{1}{X^2Y^2} \sum_{1 < q_\chi \leq w^A} d(\chi)^2 |A_{\Lambda,X}(\chi)|^2 |A_{\Lambda,Y}(\chi)|^2 = o(1),$$

because $\sum_{q|Q} d(q) = \kappa^{-1} = O(1)$.

For the complementary conductors, Chebyshev's bound gives

$$|A_{\Lambda,X}(\chi)| \leq \sum_{X < n \leq 2X} \Lambda(n) \ll X, \tag{62}$$

and likewise for Y. Hence

$$\frac{1}{X^2Y^2} \sum_{q_\chi > w^A} d(\chi)^2 |A_{\Lambda,X}(\chi)|^2 |A_{\Lambda,Y}(\chi)|^2 \ll \sum_{\substack{q|Q \\ q > w^A}} d(q) \ll e^{-A}$$

by lemma 3.2. First let $w \to \infty$ with A fixed, and then let $A \to \infty$. This proves (61). □

Let $\mathcal{L}$ be an interval of

$$H = \left\lfloor Q^{1/\log w} \right\rfloor \tag{63}$$

consecutive integers, and put $H_Q = \#\{\ell \in \mathcal{L} : (\ell, Q) = 1\}$.

Corollary 3.4 (A subpower interval of shifts)**.** *Under the hypotheses of theorem 3.3, uniformly in the location of $\mathcal{L}$,*

$$\frac{1}{XY} \left(\frac{1}{H_Q} \sum_{\substack{\ell \in \mathcal{L} \\ (\ell,Q)=1}} |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2 \right)^{1/2} = o(1). \tag{64}$$

Here $H = Q^{o(1)}$ and $H/Q \to 0$.

Proof. By proposition 2.2, the weighted deterministic completion estimate gives

$$\left| \sum_{\substack{\ell \in \mathcal{L} \\ (\ell,Q)=1}} |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2 - \frac{H}{Q} \sum_{\ell \in G} |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|^2 \right| \leq \mathfrak{C}(Q)(L_{\Lambda,X}L_{\Lambda,Y})^2,$$

where $L_{\Lambda,X} = \sum_{X < n \leq 2X, (n,Q)=1} \Lambda(n) \ll X$ and similarly $L_{\Lambda,Y} \ll Y$. Also $H_Q \sim Hg/Q$, and (43) gives $\mathfrak{C}(Q)/(H\delta_Q) = o(1)$ for the quadratic prime window. After division by $H_Q X^2 Y^2$, the completion error is $o(1)$, while the completed term is $o(1)$ by theorem 3.3. □

Corollary 3.5 (Density-one shifts in every sampled block)**.** *Under the same hypotheses, there is a deterministic sequence $\varepsilon_w \to 0$ such that, uniformly in the location of $\mathcal{L}$,*

$$\#\left\{ \substack{\ell \in \mathcal{L}:(\ell,Q)=1 \\ |\mathcal{B}_{2\ell}^{\Lambda,\Lambda}(X,Y)|>\varepsilon_w XY} \right\} = o(H_Q). \tag{65}$$

This density-one statement does not imply the estimate for any prescribed shift, and in particular it does not select $\ell = 1$.

Proof. Let S_w be the supremum, over the stated parameter range and shift blocks, of the normalized mean square in (64), and choose $\varepsilon_w = \max\{w^{-1}, S_w^{1/4}\}$. Uniformity in corollary 3.4 gives $\varepsilon_w \to 0$, while Markov's inequality bounds the exceptional proportion by $S_w/\varepsilon_w^2 \le S_w^{1/2} = o(1)$. □

3.3 A fixed-shift theorem above the product threshold

The shift average is not needed once both factor lengths are fixed positive powers of Q and their product is larger than Q by a fixed power. The additional input is the classical multiplicative large sieve, aggregated on dyadic conductor blocks against the exact multiplier $d(q)$ [9, 12].

Theorem 3.6 (Two Mangoldt weights at a prescribed shift)**.** *Fix $0 < \delta < 1$. Suppose*

$$Q^\delta \le X, Y < Q, \qquad XY \ge Q^{1+\delta}. \tag{66}$$

Then, uniformly for every h with $(h,Q) = 1$,

$$\mathcal{B}_h^{\Lambda,\Lambda}(X,Y) = o(XY). \tag{67}$$

In particular, (67) holds for the prescribed shift $h = 2$.

Proof. Choose a fixed $A > 3$ and put $W = w^A$. The characters with $q_\chi \le W$ contribute $o(XY)$ by lemma 3.1. More explicitly, there are only $w^{O_A(1)}$ divisors $q \mid Q$ below w^A, and $d(q)\#\{\chi : q_\chi = q\} = 1$ for each one. Hence the triangle inequality and (57), with a sufficiently large logarithmic saving, give

$$|\mathcal{B}_h^{q_\chi \le W}(X,Y)| \le \sum_{\substack{q \mid Q \\ 1<q\le W}} \max_{q_\chi = q} |A_{\Lambda,X}(\chi)| \max_{q_\chi=q} |A_{\Lambda,Y}(\chi)| = o(XY).$$

For the remaining conductors, use the general dyadic large-sieve tail bound (46) with $R = W$. Its coefficient energies satisfy

$$E_{\Lambda,X} := \sum_{\substack{X<n\le 2X \\ (n,Q)=1}} \Lambda(n)^2 \le \log(2X) \sum_{X<n\le 2X} \Lambda(n) \ll X \log X, \qquad E_{\Lambda,Y} \ll Y \log Y, \tag{68}$$

where the last inequality is Chebyshev's bound. Consequently,

$$\frac{|\mathcal{B}_h^{q_\chi > W}(X,Y)|}{XY} \ll \sqrt{\log X \log Y} \left\{ \frac{1}{W} + (\log Q)\left(\frac{1}{\sqrt{X}} + \frac{1}{\sqrt{Y}}\right) + \sqrt{\frac{Q}{XY}} \right\}. \tag{69}$$

The last term is $o(1)$ under (66). The W-term tends to zero because $A > 3$ and $\sqrt{\log X \log Y} \ll \log Q \asymp w^2$, while the two interval terms are killed by $X, Y \ge Q^\delta$. Together with the small-conductor estimate, this proves (67). □

3.4 Prime factors and a rough target

The fixed-shift theorem has a positive prime-supported consequence. It must not be confused with a theorem asserting that the target $pq + h$ is prime.

Corollary 3.7 (Prime factors with a locally rough shifted product)**.** *Under the hypotheses of theorem 3.6, for every fixed integer h with $(h, Q) = 1$,*

$$\sum_{\substack{p\in I_X,\ q\in I_Y\\ (pq+h,Q)=1}} \log p \log q = (\kappa + o(1))XY, \tag{70}$$

$$\#\{(p,q) \in I_X \times I_Y : (pq + h, Q) = 1\} \sim \kappa \frac{XY}{\log X \log Y}. \tag{71}$$

Both statements include $h = 2$.

Proof. Write

$$L_{\Lambda,X} = \sum_{\substack{n\in I_X\\ (n,Q)=1}} \Lambda(n), \qquad L_{\Lambda,Y} = \sum_{\substack{n\in I_Y\\ (n,Q)=1}} \Lambda(n).$$

The prime number theorem and the same prime-power deletion used in lemma 3.1 give $L_{\Lambda,X} = X + o(X)$ and $L_{\Lambda,Y} = Y + o(Y)$. Since $K_h = \kappa(1 + K_h^\circ)$ on $G \times G$, theorem 3.6 yields

$$\begin{aligned}&\sum_{\substack{m\in I_X,\ n\in I_Y\\ (mn,Q)=1,\ (mn+h,Q)=1}} \Lambda(m)\Lambda(n)\\ &\qquad = \kappa\left\{L_{\Lambda,X}L_{\Lambda,Y} + \mathcal{B}_h^{\Lambda,\Lambda}(X,Y)\right\} = (\kappa + o(1))XY.\end{aligned}$$

The total von Mangoldt weight of proper prime powers in a dyadic interval is $O(X^{1/2}(\log X)^2)$, and similarly for Y. Chebyshev's bound on the other coordinate therefore shows that all terms in which m or n is a proper prime power contribute $o(XY)$. At this point the factor primes still carry the restriction $(pq, Q) = 1$. Restoring the omitted pairs with $p \mid Q$ or $q \mid Q$ changes the logarithmically weighted sum by at most

$$\left(\sum_{\substack{r\mid Q\\ r \text{ prime}}} \log r\right)\left(\sum_{\substack{q\in I_Y\\ q \text{ prime}}} \log q\right) + \left(\sum_{\substack{p\in I_X\\ p \text{ prime}}} \log p\right)\left(\sum_{\substack{r\mid Q\\ r \text{ prime}}} \log r\right) \ll Y \log Q + X \log Q,$$

with the doubly counted intersection bounded by $(\log Q)^2$. This is $o(XY)$ because $X, Y \geq Q^\delta$. (In fact, eventually $X, Y > w^2$, so no prime in either dyadic interval divides Q.) This proves (70). Finally, $\log p = (1 + o(1)) \log X$ uniformly for $p \in I_X$, and similarly on I_Y; partial unweighting gives (71). □

Remark 3.8 (The target is rough, not prime)*.* The condition $(pq + h, Q) = 1$ only says that $pq + h$ avoids the primes in the finite window $(w, w^2]$. Neither theorem 3.6 nor corollary 3.7 inserts $\Lambda(pq + h)$, proves that $pq + h$ is prime, gives a twin-prime lower bound, or bypasses parity. Likewise, the density-one conclusion corollary 3.5 cannot be used to select the prescribed value $\ell = 1$. The threshold $XY > Q$ in the fixed-shift theorem is a range of the present large-sieve cutoff argument, not an asserted barrier.

4 Exact finite certificate

The reference implementation accompanying this paper uses only integers and rational numbers. It checks the weighted centering, the exact-conductor reconstruction, complete-shift Parseval, conductor second moments, and deterministic short-shift completion. None of these finite checks is used to infer an asymptotic statement.

Take

$$Q = 5 \cdot 7 \cdot 11 = 385, \qquad |G| = 240, \qquad \kappa = \frac{9}{16},$$

and the intervals

$$I = [19, 61], \qquad J = [47, 89].$$

On the Q-unit elements of these intervals, put

$$a_m = ((m^2 + 3m + 1) \bmod 7) - 3, \qquad b_n = ((2n + 5) \bmod 5) - 2.$$

Here the least nonnegative residue is used before the final subtraction. All coefficient sums below are restricted to the displayed interval and to Q-units. The exact coefficient statistics are

$$\sum_{\substack{m \in I \\ (m,Q)=1}} a_m = 41, \quad \|a\|_1 = 57, \quad E_a = 137, \qquad \sum_{\substack{n \in J \\ (n,Q)=1}} b_n = 16, \quad \|b\|_1 = 30, \quad E_b = 46.$$

At $h = 2$, direct counting gives

$$\sum_{\substack{m \in I,\ n \in J \\ (mn,Q)=1,\ (mn+2,Q)=1}} a_m b_n = 362,$$

and hence

$$\mathcal{B}_2(a, b) = \frac{16}{9} 362 - 41 \cdot 16 = -\frac{112}{9}. \tag{72}$$

For $1 < q \mid Q$, define the fixed-shift conductor piece and its complete-shift energy by

$$\mathcal{B}_{2,q} = \sum_{q_\chi = q} c_2(\chi) A_a(\chi) A_b(\chi), \qquad \mathcal{E}_q = \sum_{q_\chi = q} d(\chi)^2 |A_a(\chi)|^2 |A_b(\chi)|^2.$$

These exact-conductor quantities are listed in table 1.

Table 1: Weighted exact-conductor certificate at $Q = 385$.

q	5	7	11	35	55	77	385
$\mathcal{B}_{2,q}$	$-92/3$	28	$-44/9$	$-32/15$	$8/27$	$56/9$	$-1252/135$
$\mathcal{E}_q$	$6104/9$	$29228/25$	$8824/81$	$812/45$	$3584/81$	$210692/2025$	$71996/1215$

The first row sums to (72). Across all unit shifts, the exact mean is zero and

$$\frac{1}{240} \sum_{\ell \in G} |\mathcal{B}_{2\ell}(a, b)|^2 = \sum_{\substack{q \mid 385 \\ q > 1}} \mathcal{E}_q = \frac{2650976}{1215}. \tag{73}$$

For every exact conductor, the program also evaluates the integer character-orthogonality kernel

$$\prod_{p \mid q} \Big((p-1) \mathbf{1}_{x \equiv y \,(\mathrm{mod}\, p)} - 1 \Big)$$

and verifies the second-moment bounds used before large-sieve aggregation. On the shift interval $[23, 53]$, there are 20 unit shifts; the exact square-sum completion discrepancy is $199884544/31185$, below the elementary bound 895568800. These deliberately loose finite inequalities test scope and normalization, not sharp constants.

5 What has advanced and what remains open

Relative to the preceding papers in this series, the new feature is genuine arithmetic weighting on the two factor coordinates. In particular, the von Mangoldt applications are not obtained by treating Λ as a bounded perturbation of the coefficient-one problem. Small conductors use prime distribution. At complete or short shift means, the remaining conductors are controlled by the summable exact multiplier tail; at a prescribed shift, they are controlled by the multiplicative large sieve together with that multiplier.

This advance has four strict boundaries.

B1. The target is rough, not prime. The condition $(pq+h, Q) = 1$ excludes the primes in one finite window from $pq+h$. It neither asserts nor estimates the primality of $pq+h$.

B2. Factor weights are not a twin-prime correlation. The expression contains $\Lambda(m)\Lambda(n)$ on
as that correlation.

B3. Shift averaging does not select $h = 2$. The complete and subpower-shift mean-square theorems imply a density-one statement inside the averaged shift block. They do not imply that the particular member $\ell = 1$ is good. The prescribed-shift theorem is separate.

B4. The fixed-shift product threshold is methodological. For arbitrary bounded coefficients, the present large-sieve argument requires $MN/Q \to \infty$ after harmless lower bounds on the individual lengths. We prove neither an endpoint nor failure below this range.

Calling the fixed-shift estimate "Type II-like" refers only to its bilinear, coefficient-uniform form on a factor box. After decomposing a target-prime weight such as $\Lambda(mn+h)$, a genuine Type II term would carry additional convolution geometry. Nothing here estimates such a term.

5.1 The next precise gap

Within the present framework, one next question is to cross the ambient-volume threshold by proving a fixed-shift estimate on boxes with

$$M = Q^{\lambda+o(1)}, \qquad N = Q^{\nu+o(1)}, \qquad \lambda + \nu \leq 1.$$

The proof in this paper loses the phase of each conductor block through Cauchy–Schwarz and pays the D^2 term in the multiplicative large sieve. Breaking that threshold would require additional dispersion, cancellation between conductor blocks, or structure in the coefficients. Even a successful estimate of this kind would still leave the independent problem of placing a prime-sensitive weight on $mn+h$ and obtaining a positive main term despite the parity obstruction.

Conversely, a rigorous example showing saturation of the conductor-block estimate for a proposed coefficient class would be a useful negative result: it would identify which version of the large-sieve route cannot reach the next scale. The paper therefore supplies a falsifiable analytic interface rather than a claimed route around parity.

6 Conclusion

The exact conductor decomposition of a centered CRT kernel remains effective after arithmetic weights are introduced. Character orthogonality gives coefficient-uniform shift means, the multiplicative large sieve gives a fixed-shift estimate above the ambient-volume threshold $MN = Q$, and Siegel–Walfisz permits von Mangoldt weights to pass through the small-conductor range. This yields both power-short prime-weighted shift means and a prescribed-shift prime-factor/rough-target asymptotic.

These are strict local advances. The target value is not proved prime, the fixed shift below the product threshold is not controlled, and parity is untouched. The value of the result is the separation it enforces: factor-prime distribution, conductor aggregation, prescribed-shift cancellation, target-prime detection, and positivity are distinct interfaces and must be proved separately.

Data and code availability

The manuscript source, exact rational certificate, and standard-library Python tests are included in the repository directory for this paper. From the `experiments` directory, running

```
python exact_weighted_diagnostics.py --output data/exact-certificate.json
python -m unittest -v test_exact_weighted_diagnostics.py
```

regenerates the certificate and verifies the finite identities. No random seed, floating-point Fourier transform, or external data set is used.

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